

Integrator

The HiRep code uses a multilevel integrator and each integrator level has to be specified in the input file.

```
integrator {  
    level = 0  
    type = o2mn  
    steps = 10  
}
```

Variable	Description
level	unique integrator level (level 0 is the outermost level)
type	integrator type (see below)
steps	number of integration steps

The table below show the different integrators implemented in the HiRep code. The last column in the table show how many times the next integrator level will be called in each iteration of the given integrator.

Type	Description	Next level calls
1f	leap-frog integrator	1
o2mn	2nd order minimal norm (omelyan) integrator	2
o4mn	4th order minimal norm (omelyan) integrator	5

Gauge

The gauge monomial is the standard Wilson plaquette action.

$$S = -\frac{\beta}{N} \sum_{x, \mu > \nu} \text{Re tr}(U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x)) \quad (1)$$

The example below show how to specify a gauge monomial in the input file.

```
monomial {  
    id = 0  
    type = gauge  
    beta = 2.0  
    level = 0  
}
```

Variable	Description
id	unique monomial id
type	monomial type
beta	bare coupling for the gauge field
level	integrator level where the monomial force is evaluated

HMC

The HMC monomial is the standard term for simulating two mass degenerate fermions.

$$S = \phi^\dagger (D^\dagger D)^{-1} \phi \quad (2)$$

The example below show how to specify an HMC monomial in the input file.

```
monomial {  
    id = 1  
    type = hmc  
    mass = -0.750  
    mt_prec = 1e-14  
    force_prec = 1e-14  
    mre_past = 5  
    level = 1  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
mt_prec	inverter precision used in the Metropolis test
force_prec	inverter precision used when calculating the force
mre_past	number of past solutions used in the chronological inverter
level	integrator level where the monomial force is evaluated

Comments

- When using the chronological inverter the force precision should be 10^{-14} or better to ensure reversibility in the algorithm.

TM

In this monomial the twisted mass is added before the Dirac operator has been even/odd preconditioned. The example below shows how to specify the monomial in the input file.

```
monomial {  
  id = 1  
  type = tm  
  mass = -0.750  
  mu = 0.1  
  mt_prec = 1e-14  
  force_prec = 1e-14  
  mre_past = 5  
  level = 1  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
mu	bare twisted mass
mt_prec	inverter precision used in the Metropolis test
force_prec	inverter precision used when calculating the force
mre_past	number of past solutions used in the chronological inverter
level	integrator level where the monomial force is evaluated

Comments

- When using the chronological inverter the force precision should be 10^{-14} or better to ensure reversibility in the algorithm.

TM Alternative

In this monomial the twisted mass is added after the Dirac operator has been even/odd preconditioned. The example below shows how to specify the monomial in the input file.

```
monomial {  
  id = 1  
  type = tm_alt  
  mass = -0.750  
  mu = 0.1  
  mt_prec = 1e-14  
  force_prec = 1e-14  
  mre_past = 5  
  level = 1  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
mu	bare twisted mass
mt_prec	inverter precision used in the Metropolis test
force_prec	inverter precision used when calculating the force
mre_past	number of past solutions used in the chronological inverter
level	integrator level where the monomial force is evaluated

Comments

- When using the chronological inverter the force precision should be 10^{-14} or better to ensure reversibility in the algorithm.

Hasenbusch

The hasenbusch term is a mass preconditioned term, used in connection with an HMC monomial.

$$S = \phi^\dagger \left(\frac{D^\dagger D}{(D + \Delta m)^\dagger (D + \Delta m)} \right) \phi \quad (3)$$

The example below show how to specify a Hasenbusch monomial in the input file.

```
monomial {  
    id = 1  
    type = hasenbusch  
    mass = -0.750  
    dm = 0.1  
    mt_prec = 1e-14  
    force_prec = 1e-14  
    mre_past = 2  
    level = 0  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
dm	shift in the bare mass
mt_prec	inverter precision used in the Metropolis test
force_prec	inverter precision used when calculating the force
mre_past	number of past solutions used in the chronological inverter
level	integrator level where the monomial force is evaluated

Comments

- When using the chronological inverter the force precision should be 10^{-14} or better to ensure reversibility in the algorithm.
- The mass used in the associated HMC monomial should be the mass of this monomial plus the mass shift.

TM Hasenbusch

The example below show how to specify a Hasenbusch monomial with even odd preconditioned twisted mass

```
monomial {  
    id = 1  
    type = hasenbusch_tm  
    mass = -0.750  
    mu = 0  
    dmu = 0.1  
    mt_prec = 1e-14  
    force_prec = 1e-14  
    mre_past = 2  
    level = 0  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
mu	twisted mass
dmu	shift in the twisted mass
mt_prec	inverter precision used in the Metropolis test
force_prec	inverter precision used when calculating the force
mre_past	number of past solutions used in the chronological inverter
level	integrator level where the monomial force is evaluated

Comments

- When using the chronological inverter the force precision should be 10^{-14} or better to ensure reversibility in the algorithm.
- The twisted mass μ used in the associated TM monomial should be the twisted mass of this monomial plus the twisted mass shift.

TM Hasenbusch Alternative

The example below show how to specify a Hasenbusch monomial with twisted even odd preconditioned operator.

```
monomial {  
    id = 1  
    type = hasenbusch_tm_alt  
    mass = -0.750  
    mu = 0  
    dmu = 0.1  
    mt_prec = 1e-14  
    force_prec = 1e-14  
    mre_past = 2  
    level = 0  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
mu	bare twisted mass
dmu	shift in the twisted mass
mt_prec	inverter precision used in the Metropolis test
force_prec	inverter precision used when calculating the force
mre_past	number of past solutions used in the chronological inverter
level	integrator level where the monomial force is evaluated

Comments

- When using the chronological inverter the force precision should be 10^{-14} or better to ensure reversibility in the algorithm.
- The twisted mass μ used in the associated TM monomial should be the twisted mass of this monomial plus the twisted mass shift.

RHMC

The RHMC monomial uses a rational approximation to simulate an odd number of mass degenerate fermions.

$$S = \phi^\dagger (D^\dagger D)^{-n/d} \phi \quad (4)$$

The example below show how to specify an RHMC monomial in the input file.

```
monomial {  
  id = 1  
  type = rhmc  
  mass = -0.750  
  n = 1  
  d = 2  
  mt_prec = 1e-14  
  md_prec = 1e-14  
  force_prec = 1e-14  
  level = 0  
}
```

Variable	Description
id	unique monomial id
type	monomial type
mass	bare fermion mass
n	fraction numerator
d	fraction denominator
mt_prec	inverter precision used in the Metropolis test
md_prec	precision of the rational approximation
force_prec	inverter precision used when calculating the force
level	integrator level where the monomial force is evaluated